

## **Session 1: Python Introduction**

Introduction & Installation in Python

## **Session 2: Input & Output**

Input & Output, Variables, Data Types, Type Conversion

## **Session 3: Basic Data Types**

int, float, string, Boolean, Complex, List, Tuple, Dictionary, Set

## **Session 4: Functions**

Types of Functions, Function with Arguments, Recursion, Global, Local and Non-local, Lambda Functions

## **Session 5: Built-in Methods**

Understanding Built-in Methods: Math, String, and Date Functions, Operations, PDF Extraction, CSV Module

## **Session 6: Flow Control**

Operators, Conditional Statements (if, if-else, if-elif-else, Nested if), Loops (for, for with range, for-else, while), break and continue

## **Session 7: Modules and Packages**

Creating Module, Using Module, Creating Package, Importing Modules from Package

## **Session 8: File Operations**

open(), write(), writelines(), read(), readline(), readlines(), close(), File Pointer Manipulation

## **Session 9: Directories**

Current Working Directory, Change Directory, List Directory, Create Directory, Rename Directory, Delete Directory or File

## **Session 10: Exception Handling**

Try, Except, Finally, Try-Else, Custom Exception, Error vs Exception

## **Session 11: OOP Concepts**

Class and Objects, Inheritance, Encapsulation, Polymorphism, Data Abstraction, Overloading, Overriding

## **Session 12: Multithreading**

Understanding Threads, Forking Threads, Synchronization

## **Session 13: Database Connectivity**

MySQL and MongoDB, CRUD Operations, Queries

## **Session 14: CGI Applications**

Introduction and Writing Python Programs for CGI

### **Session 15: Regular Expressions**

Pattern Matching, Searching, Real-time Parsing, Validation Concepts

### **Session 16: SMTP and Email Handling**

Sending Emails, Sending Attachments

### **Session 17: Real-Time Applications**

Web Scraping, Chatbot, WhatsApp Chatbot, Language Translation, Image Resolution Handling, Similarity Ratio, Sentence Tagging, QR Code Generator, Spell Checker, Wikipedia Scraping, Anagram